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Chaos and DNA Blended Hybrid Encryption Algorithm for Secure Image Transmission over DCT Pre-coded OFDM

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Abstract
In this paper, a hybrid security solution based on the chaotic map for a discrete cosine transform (DCT) precoded orthogonal frequency division multiplexing (OFDM) system is proposed. The two-level encryption scheme is employed to encrypt the image in the upper and physical layer of the OFDM system. At first, bit-level encryption based on Deoxyribo Nucleic Acid coding and combined chaotic map is done in the upper layer of the proposed system. Then, the encrypted bitstreams are fed to the OFDM transmitter where the real and imaginary parts of the quadrature amplitude modulated symbols are scrambled chaotically to achieve the symbol level encryption in the physical layer of OFDM system. In addition to the hybrid security model, DCT precoding is added to achieve the trade-off between bit error rate (BER) and peak to average power ratio (PAPR) performance. Experiments are carried out to validate the strength of encryption algorithms against key sensitivity, statistical and differential attacks. Further, the BER, PAPR and peak signal to noise ratio performances have been analyzed for various cases to prove the strength of the proposed system to achieve enhanced security with better power and BER performance.

Keywords OFDM · DCT precoding · PAPR · Combined chaotic map · Upper and physical layer encryption

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1 Introduction

Orthogonal frequency division multiplexing (OFDM) is a reliable and robust modulation system used for transmission of digital data over a radio channel. This system has gained increasing popularity in various wireless communication systems because of its capability to provide higher data rates and robustness to multipath effects. In the past few decades, the requirement of such a high-speed wireless communication system has increased intensely. Advancements in the fields of computer networks, satellite applications, telemedicine applications and other Internet applications have provoked the need for proper security and privacy framework in the wireless communication systems. Thus, the secure transmission of digital data through the wireless channel has become a pivotal research topic.

Initially, the security of the communication system was based on upper-layer security schemes such as an end to end encryption, and access control schemes. Recently, to boost the security in wireless systems, Physical Layer Security (PLS) schemes has ensued as an alternate security solution. This type of PLS is first used in the radio frequency-based OFDM systems [1, 2]. A remarkable survey on physical layer security in OFDM systems is reported in [3]. Few notable existing solutions for PLS focus mainly on developing the pseudo-random number key generation [4], authentication [5], artificial noise-based security [6], channel-based encryption [7], etc. Apart from these solutions, Physical Layer Encryption (PLE) is introduced as another approach to achieve the secure transmission of digital data in the wireless medium. This kind of PLE is different from PLS in the fact that the PLE operates in the physical layer of the OFDM system. Many existing works are available with the literature with PLE [8–10]. Ruifeng et al. have proposed a secure OFDM system based on PLE [8]. This work is based on constellation rotation in the pseudo-random fashion, and the system security is enhanced by inserting the artificial noise. The system has reported good security, but the system has suffered from the Bit Error Rate (BER) problem. Wei et al. [9] have proposed an asymmetric PLE based OFDM system. Asymmetric PLE based on Elliptic Curve Cryptography (ECC) has reduced the problem of key distribution with good BER and better constellation entropy. In [10], a PLE system based on subcarrier obfuscation and symbol scrambling has proposed to enhance the security of the OFDM system. The results have reported that the system has better performance in terms of search space and complexity.

The main limitation of the OFDM system is that it severely suffers from high Peak to Average Power Ratio (PAPR), which degrades the power and BER performance. Many PAPR reduction techniques based on linear and non-linear companding [11–13], precoding [14], clipping [15], carrier reduction [16] have been reported in the literature. Among the suggested PAPR reduction techniques, precoding is one of the efficient and accessible techniques [17, 18]. Thus, combining PLE with precoding techniques may solve the existing problem of security and power efficiency. Furthermore, the combination of physical and Upper Layer Encryption (ULE) provide better security solution in and out of the OFDM system. The chaos and DNA based security schemes have attracted many researchers in the past two decades because of its unusual feature to generate the pseudo-random sequences and enhanced diffusion property, respectively [19]. The DNA and chaos-based encryption schemes have been investigated in our previous works [20–23].

Understanding the astonishing feature of chaotic maps, few researchers have utilized the chaotic maps for confusing and diffusing the OFDM signal matrix. Some of the works are presented below. Hu et al. have proposed a PLE in OFDM system based on 4-D hyperchaos to enhance the confidentiality of the signal in the physical layer [24]. The system has achieved

keyspace of 10^{120}, but the usage of hyperchaos has increased the computational overhead of the system. Liu et al. and Zhang et al. have suggested the chaotic permutation based on single-dimensional chaotic maps in the physical layer. A PLE with 3-dimensional Brownian motion and chaos to confuse and diffuse the OFDM signal is presented in [25]. Both the systems have experienced lesser computational complexity than Hu et al. system [25–27]. Multilevel chaotic encryption is carried out by Wei et al. [28]. The OFDM signal is encrypted in four different stages to achieve 22.06 Gb/s transmission rate. Though the system seems to be secure, it suffered from a high PAPR rate of 14.5 dB [28]. Wu et al. have proposed an optimum block division based on 2D Logistic Adjusted Sine Map (LASM) to divide the OFDM signal matrix. The system has achieved 3 dB PAPR reduction with 22.06 Gbps transmission rate. This method is highly efficient with less computational complexity with good PAPR reduction and reduced power [29].

Based on the above analysis, the proposed work presents the hybrid security solution based on DNA-chaotic schemes in the Discrete Cosine Transform (DCT) precoded OFDM system. The security part of the proposed works fuses the bit-level encryption in ULE with symbol-level scrambling in PLE to enhance the security. The main contributions of the proposed study are summarized as follows:

(1) The hybrid two-level encryption in the upper and physical layer enhances the security of the system
(2) Two-level encryption scheme not only enhances the security but also helpful in reducing the PAPR
(3) PAPR and BER performance of the OFDM system in the presence of PLE and DCT precoding is studied
(4) In ULE and PLE, instead of 1-D chaotic schemes, combined chaotic maps are used to increase the keyspace and sensitivity
(5) The resistance of the proposed encryption scheme towards statistical and differential attacks are studied

In this view, the paper is organized as follows. Section 2 of the manuscript gives the introduction about combined chaotic system and DNA coding techniques used in ULE and PLE of the proposed system. System design is discussed in Sect. 3, experimental results and discussion on the results are carried out in Sects. 4, 5 concludes the paper.

2 Preliminaries

2.1 Combined Chaotic Map (CCM)

The system is employing combined chaotic maps instead of a single 1-D chaotic map because of the advantages studied in our previous research works [20, 21]. Three 1-D chaotic maps namely logistic, tent and sine maps are utilized in the proposed scheme for generating the pseudo-random sequences. A combined chaotic map is developed by combining logistic-tent, logistic-sine and tent-sine in the single system. Let P(f1,g1) and Q(f2,g2) represents the 1D chaotic maps, where f1, f2 represents the initial seeds and g1, g2 represents the control parameters of P and Q respectively. The mathematical expression of the developed combined chaotic map is $X_{CCM} = \text{mod}\ (P(f1,g1) + Q(f2,g2),1)$[23]. The mathematical representations

of Combined Logistic-Sine (CLS), Combined Logistic-Tent (CLT) and Combined Tent-Sine (CTS) are,

$$R_{n+1} = \left(\gamma sin\left(\pi * R_n\right)/4 + \rho * R_n\left(1 - R_n\right)\right) \bmod 1 \tag{1}$$

$$R_{n+1} = \begin{cases} \left(\frac{\mu R_n}{2} + \rho * R_n\left(1 - R_n\right)\right) mod 1; & for\, R_n < 1/2 \\ \left(\frac{\mu\left(1-R_n\right)}{2} + \rho * R_n\left(1 - R_n\right)\right) mod 1; & for\, R_n \geq 1/2 \end{cases} \tag{2}$$

$$R_{n+1} = \begin{cases} \left(\frac{\mu R_n}{2} + \gamma sin\left(\pi * R_n\right)/4\right) mod\, 1; & for\, R_n < 1/2 \\ \left(\frac{\mu\left(1-R_n\right)}{2} + \gamma sin\left(\pi * R_n\right)/4\right) mod\, 1; & for\, R_n \geq 1/2 \end{cases} \tag{3}$$

where R_n is the initial seed with range (0,1) and ρ, γ, μ are the control parameters. All the three systems behave chaotically in the entire interval of [0, 4]. Figure 1a–f illustrates the chaotic range achieved with logistic, tent, sine, CLT, CLS, CTS, respectively. From Fig. 1, it can be understood that combining two 1D maps in a single system may escalate the chaotic range, which ultimately increases the randomness, keyspace and sensitivity.

The higher positive value of Lyapunov exponents specifies quicker divergences, and so assures better chaotic behaviour. The Lyapunov exponent plot of logistic, tent, sine, CLT, CLS and CTS maps are shown in Fig. 2. From Fig. 2, it can be observed that the Lyapunov exponent values of CLS, CLT and CTS are higher than that of the seed maps. This again proves that the maps derived from CCM can achieve enhanced chaotic behaviour and consequently have excellent randomness and well suited for information security applications.

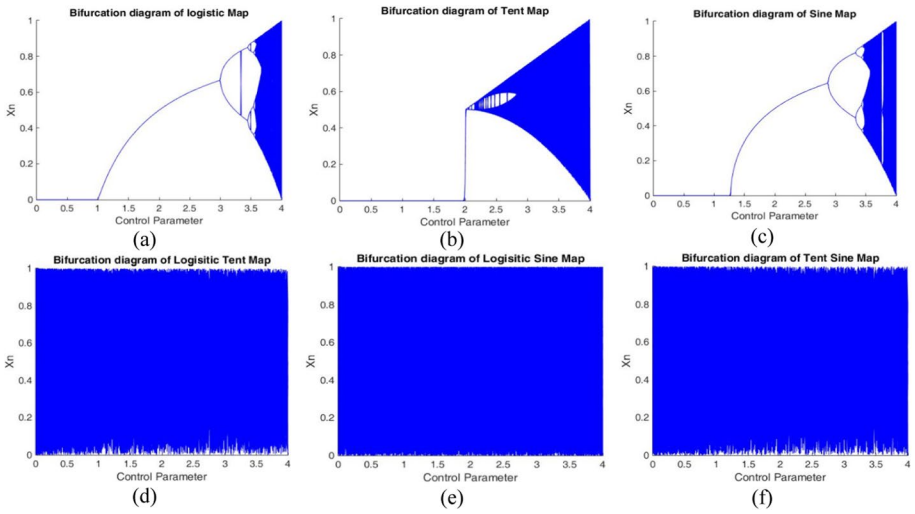


Fig. 1 Bifurcations diagram: **a** Logistic map, **b** Tent Map, **c** Sine map, **d** CLT map, **e** CLS maps, and **f** CTS map

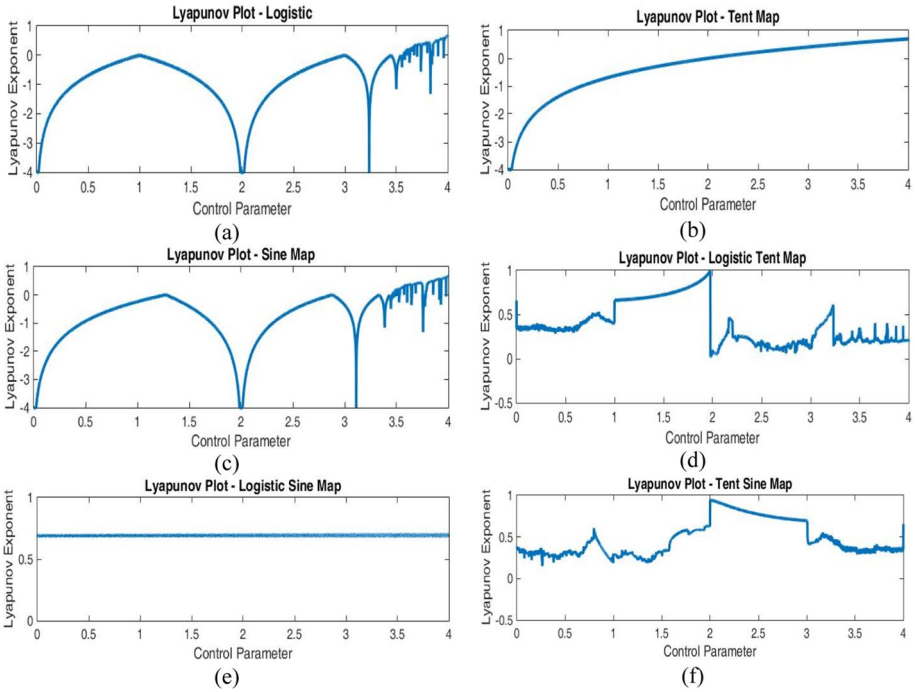


Fig. 2 Lyapunov exponents: **a** Logistic map, **b** Tent Map, **c** Sine map, **d** CLT map, **e** CLS maps, and **f** CTS map

2.2 DNA encoding

DNA consists of 4 building blocks: Adenine (A), Guanine (G), Cytosine (C) and Thymine (T). According to Watson-Crick's rule, A and T are complementary to each other as are C and G. In the binary system, 0 and 1 are complementary to each other. Similarly, 00, 11 and 01, 10 are complementary to each other. Considering this criterion,

Table 1 DNA encoding rules

	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6	Rule 7	Rule 8
00	A	A	T	T	G	G	C	C
01	G	C	G	C	A	T	A	T
10	C	G	C	G	T	A	T	A
11	T	T	A	A	C	C	G	G

Table 2 DNA XOR rules

XOR	A	C	G	T
A	A	C	G	T
C	C	A	T	G
G	G	T	A	C
T	T	G	C	A

totally 8 encoding rules can be derived which satisfy the Watson-Crick rule. The DNA encoding rules that suit practical DNA computing are listed in Table 1. Table 2 shows the XOR rules followed in DNA operation. The DNA XOR operation in Table 2 is reflexive.

3 System Design

The main focus of the proposed scheme is secure transmission of digital images over DCT precoded OFDM system. The overall block diagram of the proposed architecture is portrayed in Fig. 3.

Secure image transmission is achieved by performing two levels of encryption, in which the first level encryption is done in the upper layer and the second level of encryption is carried out in the physical layer of the system.

3.1 Upper Layer Encryption

This section covers the first level of encryption, i.e., encrypting the digital image in the upper layer of the DCT precoded OFDM system. The encryption algorithm is given as follows:

ALGORITHM 1 Upper layer encryption

Input: *Plain image P, Initial Seed* R_0, R_0' *and Control Parameter* γ_1, ρ_1,μ_2, ρ_2

Output: *Cipher image C*

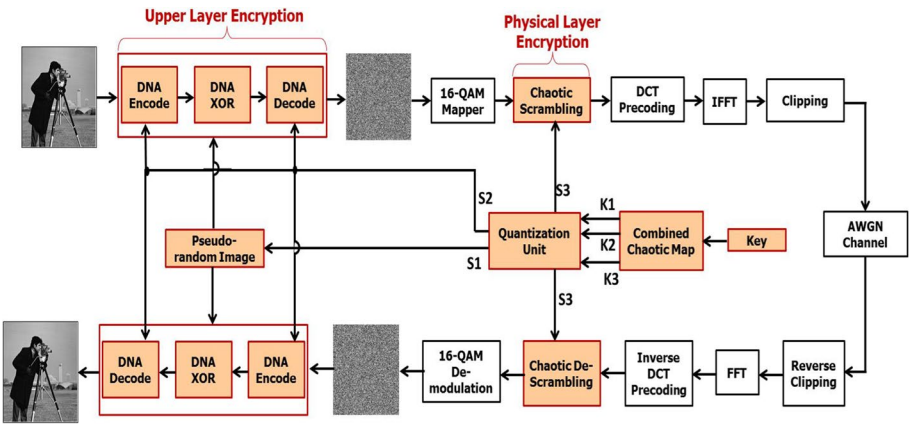


Fig. 3 Overall block diagram of the proposed architecture

Step 1 Input the plain image P of size M×N. The plain image P is transformed into the 2-D array $P=p(i, j)_{M\times N}$, where i∈(1, 2, 3,… M), j∈(1, 2, 3,…N) and p(i, j) represents a pixel of the image 'P' at ith row and jth column.

Step 2 Feed the initial seed $R_0$and control parameter γ_1 and ρ_1 to the CLS map as in Eq. (1). Let 't' be the user-defined value. To avoid the transient effect, iterate the CLS map for t times. After 't' iterations continue the iteration for (M*N) times to get the chaotic sequence. Chaotic sequence is generated as in Eq. (4),

$$K_1 = \left\{k_{t+1}, k_{t+2}, \ldots, k_{(t+(M*N)+3)}\right\},\ K_1 = \left\{k_{t+Z},\ Z \in [1, (M*N)]\right\} \tag{4}$$

Step 3 Feed the generated chaotic sequence K_1 to the quantization unit to get the quantized pseudo-random matrix S_1 as follows,

$$S_1 = \left(\text{floor}\left(K_1 \times 10^{14}\right)\right) \bmod 256 + 1,\ K_1 \in \left\{K_1 = k_Z,\ Z \in [1, (M*N)]\right\} \tag{5}$$

Step 4 A 1-D pseudo-random array of size M*N is obtained by following the steps 2 and 3. The obtained array is reshaped into a 2-D array 'R' of size M×N, same as the size of plain image P.

Step 5 Similar to step 2, Feed the initial seed R_0' and control parameter μ_2, $\rho_2$to the CLT map as in Eq. (2). Encode the pixels of plain image P and pseudo-random array R using the set of rules presented in Table 1. The rules are selected from the chaotic sequences S_2 as in Eq. (6),

$$S_2 = \left(\text{floor}\left(K_2 \times 10^{14}\right)\right) \bmod 8 + 1,\ K_2 = \left\{k_{t+1}, k_{t+2}, k_{(t+3)}\right\} \tag{6}$$

Using the first element of S_2, the pixels of P is encoded into the equivalent DNA codes. They were similarly using the second element of S_2, the elements of Rare DNA coded. Let P_E and R_E be the DNA encoded plain image and pseudo-random matrix.

Step 6 Perform DNA XOR operation on P_E and R_E using Table 2. The resultant diffused DNA matrix is 'D'.

Step 7 Using the last element of S_2, the elements of matrix D is decoded into the pixels of the range [1, 255] using Table 1. This final step gives the final cipher image C.

3.2 Physical Layer Encryption

Before entering the OFDM transmission system, the cipher image from level-1 encryption is converted into the bitstreams. Then the bits are modulated using the Quadrature Amplitude Modulation (QAM) technique, where the bits are mapped into QAM symbols. The QAM symbols are scrambled using the chaotic sequences generated from the CTS map as described in algorithm 2.

ALGORITHM 2: *Physical layer encryption/Chaotic Interleaving*

Input: *QAM symbols, Initial Seed R_0'' and Control Parameter μ_3, γ_3, User-defined number t1*

Output: *Encrypted symbols.*

Step 1 Input the QAM symbols of length M_0. $Q = [Q_1, Q_2, \ldots, Q_{Mo}]$.

Step 2 Feed the initial seed R_0'' and control parameter μ_3, $\gamma_3$to the CTS map as in Eq. (3). Iterate the sequence for 't1' times. Continue the iteration for extra M_0 times to get the required set of pseudo-random numbers. Generated chaotic sequence is,

$$K_3 = \left\{k_{t1+1}, k_{t1+2}, \ldots, k_{(t1+M0)}\right\}, \; K_3 = \left\{k_{t1+Z'}, Z' \in \left[1, M_0\right]\right\} \tag{7}$$

Step 3 Sort the generated chaotic sequence K_3 in ascending order K_3 and map the weights of sorted sequence S_3 to the QAM symbols. According to the weights of K_3, the QAM symbols are scrambled. By doing so, additional interleaving effect has been achieved among the QAM symbols; thus, the burst errors may be controlled.

Step 4 The real and imaginary part of the encrypted QAM signal can be represented as in Eq. (8)

$$Q'_m = \text{real}\left(Q'_m\right) + \text{j.imag}\left(Q'_m\right) \tag{8}$$

Since CLT map is utilized to scramble the QAM symbols in the physical layer of the OFDM system, the image transmission is highly secure with huge keyspace.

3.3 DCT Precoded Chaotic Interleaved OFDM System

First, the plain image 'P' is encrypted using the CCM and DNA coding techniques to get the cipher image 'C'. The cipher image is then converted into the bitstreams, which are then mapped into m-QAM symbols. These m-QAM symbols are interleaved or scrambled using the sorted chaotic sequences to get the final cipher bitstreams. The physical layer encrypted signal vector can be represented as in Eq. (9),

$$Q' = \left[Q'_1, Q'_1, \ldots Q'_{mo}\right] \tag{9}$$

DCT precoding is done on the signal vector Q' to generate the precoded signal vector, $Y = PQ'$, where P represents the precoding matrix. The precoded signal vector can be represented as in Eq. (10),

$$Y = \left[Y_1, Y_2, \ldots Y_{mo}\right] \tag{10}$$

The DCT precoded signal vector is a complex-valued frequency-domain symbol vector which can be converted into real-valued time-domain OFDM signal vector by applying the Inverse Fast Fourier Transform (IFFT) on Y.

In order to achieve this, the input to the IFFT unit should satisfy Hermitian symmetry. The Hermitian symmetry can be achieved by doing the operation specified in Eq. (11),

$$\begin{aligned} \mathrm{A} &= \left[0\ \mathrm{Y}_1 \ldots\ \mathrm{Y}_{\mathrm{M0}} 0\ \mathrm{Y}_{\mathrm{M0}} *\ \ldots\ \mathrm{Y}_1 *\right] \\ \mathrm{A}_{\mathrm{K}} &= \mathrm{Y}_{\mathrm{K}},\ \mathrm{K} = 2,,3,\ldots \mathrm{M}_0 + 1 \end{aligned} \tag{11}$$

The real-valued time-domain OFDM signal $\{a_n\}$ from IFFT block can be expressed as in Eq. (12),

$$a_n = \frac{1}{N_0} \sum_{K=1}^{N_0-1} \left(A_K \exp \frac{j2\pi kn}{N_0} \right), \tag{12}$$

where N_0 is the number of subcarriers.

At the receiver end, the time domain signal $\{a_n\}$ is transformed into the frequency domain signal by feeding the received time-domain signal to the FFT unit. The frequency-domain signal can be represented as in Eq. (13),

$$\begin{aligned} F_K &= \sum_{K=1}^{N_0-1} \left(a_n \exp \frac{-j2\pi kn}{\mathrm{N}} \right), \quad 0 \le K \le N_o - 1 \\ F_K &= H_K . A_K + W_K \end{aligned} \tag{13}$$

where W_K represents frequency domain additive complex Gaussian noise, H_K represents channel gain at the subcarrier K.

Let A_{KP} be the pilot symbol at K_p position, the frequency gain can be estimated as in Eq. (14),

$$\widehat{H_{KP}} = \frac{F_{KP}}{A_{KP}}, \quad \mathrm{K_p} = 1, 2, \ldots\ \mathrm{N_p} \tag{14}$$

Based on this, the frequency response of the data sub carrier can be estimated by utilizing the interpolation technique. The data symbol at K^{th} sub carrier is given in Eq. (15),

$$\widehat{A_K} = A_K + \frac{\widehat{H_K}^*}{(\widehat{H_K})^2} \cdot W_K, \widehat{Y_K} = \widehat{A_K}, \quad \mathrm{K} = 2, 3, \ldots\ \mathrm{M}_0 + 1 \tag{15}$$

The signal vector is then inverse DCT pre-coded by using the inverse precoding matrix P^H to get $\widehat{Q'}$,

$$\hat{Q}' = \mathrm{P}^{\mathrm{H}} \cdot \hat{Y} \tag{16}$$

The signal vector $\widehat{Q'}$ is then de-interleaved as in algorithm 2 to get the QAM symbols Q. After QAM demodulation, the real and complex part of the signal vector is converted into the bitstreams. These bitstreams are then decrypted by applying the reverse of algorithm 1 to get the deciphered plain image at the receiver end.

4 Results and Discussion

This section discussed the effectiveness of the proposed scheme with various analysis, namely, encryption quality analysis, communication channel analysis, PAPR and BER analysis. Experiments are carried out with five gray scale and five DICOM test images of size 256×256 in MATLAB R2015a platform.

Table 3 Comparison of keyspace with the available methods

Existing methods	Keyspace
Hu et al. [24]	10^{120}
Liu et al. [25]	10^{111}
Zhang et al. [26]	10^{90}
Wang et al. [17]	10^{90}
Proposed	10^{126}

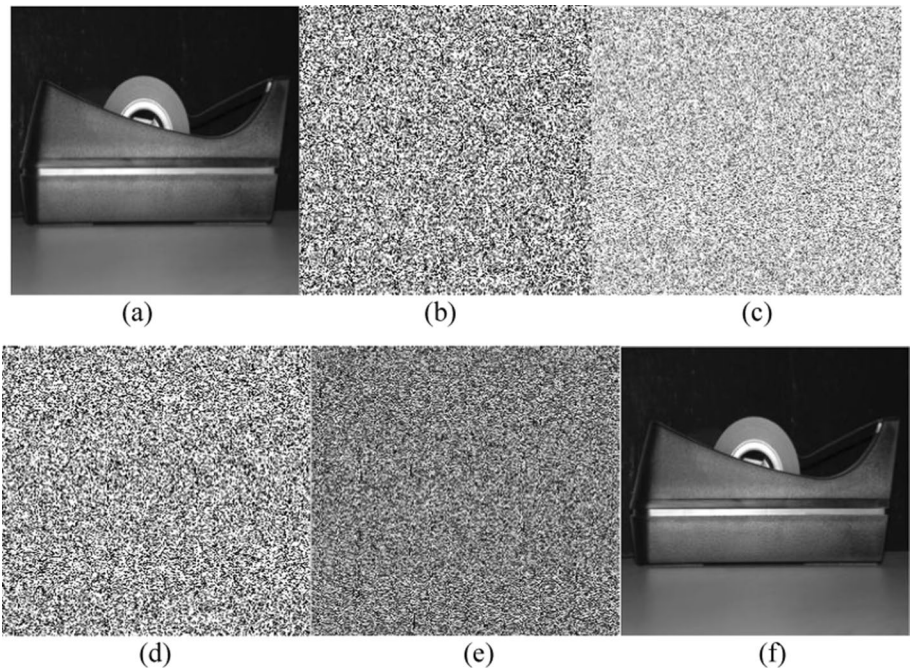


Fig. 4 Key sensitivity analysis **a** Original image, **b** Encrypted cipher image with key set K, **c** ecrypted image with key set Kw^1, **d** Decrypted image with key set Kw^2, **e** Decrypted image with key set Kw^3, **f** Decrypted image with correct key set K

4.1 Encryption Quality Analysis

This section analyses and discusses the quality of the proposed image encryption algorithm in terms of the keyspace, key sensitivity, and resistance towards statistical and differential attacks.

4.1.1 Key Space Analysis

The strength of the encryption algorithm lies in the strength of the key. A good encryption algorithm should have keyspace more substantial than 2^{128} to resist brute-force attack. The proposed scheme utilizes three key sets to encrypt the image in the upper and physical

layer of the DCT precoded OFDM system. Each key set, in turn, comprises three keys, i.e., one initial seed and two control parameters. In total, nine keys are employed in the proposed encryption model. If the key precision is set to 14, then the total keyspace of the scheme is $10^{(14\times9)}=10^{126}\approx2^{420}$. The keyspace of the proposed scheme is larger than the existing schemes as listed in Table 3 and is huge enough to resist brute-force attack.

4.1.2 Key Sensitivity Analysis

This analysis is one the most important analysis to prove the sensitivity of encryption algorithm towards the change in key. The cryptosystem is proven to be effective if the single bit change in the keyset tends to produce a different cipher image. The key sensitivity analysis of the proposed image encryption scheme is portrayed in Fig. 4. Figure 4b is the encrypted image of 4a with the key set $K=\{R_0'=0.65438798765476,\ \gamma_1=2.78768787897987,\ \rho_1=3.64564564564564,\ R_0''=0.56834509834527,\ \mu_2=3.45723865423876,\ \rho_2=2.87654312342398,\ R_0''=0.12653765762314,\ \mu_3=3.7894539547809,\ \gamma_3=3.23459812293425\}$. The key sensitivity analysis is carried out by changing the single bit in the original key set K. Three wrong key sets Kw^1, Kw^2, Kw^3 have been developed for this purpose.

(1) $Kw.^1=\{R_0'=0.65438798765476,\ \gamma_1=2.7876\mathbf{9}787897987,\ \rho_1=3.64564564564564,\ R_0''=0.56834509834527,\ \mu_2=3.45723865423876,\ \rho_2=2.87654312342398,\ R_0''=0.12653765762314,\ \mu_3=3.7894539547809,\ \gamma_3=3.23459812293425\}$

(2) $Kw.^2=\{R_0'=0.65438798765476,\ \gamma_1=2.78768787897987,\ \rho_1=3.64564564564564,\ R_0''=0.56834509834527,\ \mu_2=3.4572384\mathbf{5}423876,\ \rho_2=2.87654312342398,\ R_0''=0.12653765762314,\ \mu_3=3.7894539547809,\ \gamma_3=3.23459812293425\}$

(3) $Kw.^3=\{R_0'=0.65438798765476,\ \gamma_1=2.78768787897987,\ \rho_1=3.64564564564564,\ R_0''=0.56834509834527,\ \mu_2=3.45723865423876,\ \rho_2=2.87654312342398,\ R_0''=0.12653765762314,\ \mu_3=3.7894539547809,\ \gamma_3=3.2345981\mathbf{2}393425\}$

Figure 4c–e are the deciphered images obtained from the cipher image when decrypted using wrong key sets Kw^1, Kw^2, Kw^3respectively. Finally, Fig. 4f shows the deciphered image by using the correct key set K. From the Fig. 4; it can be inferred that the single bit change in the original key results in a different decrypted image, which proves the efficiency of the proposed algorithm towards key. Further, the analysis is validated by computing the difference between the cipher images decrypted with different keys. The computed differences are tabulated in Table 4. The results in Table 4 again prove that the proposed algorithm is highly sensitive to key.

Table 4 Difference value estimation for key sensitivity analysis

Difference b/w	4(f)–4(c)	4(f)–4(d)	4(f)–4(e)	4(c)–4(d)	4(c)–4(e)	4(d)–4(e)
Differences	99.61%	99.64%	99.63%	99.63%	99.59%	99.63%

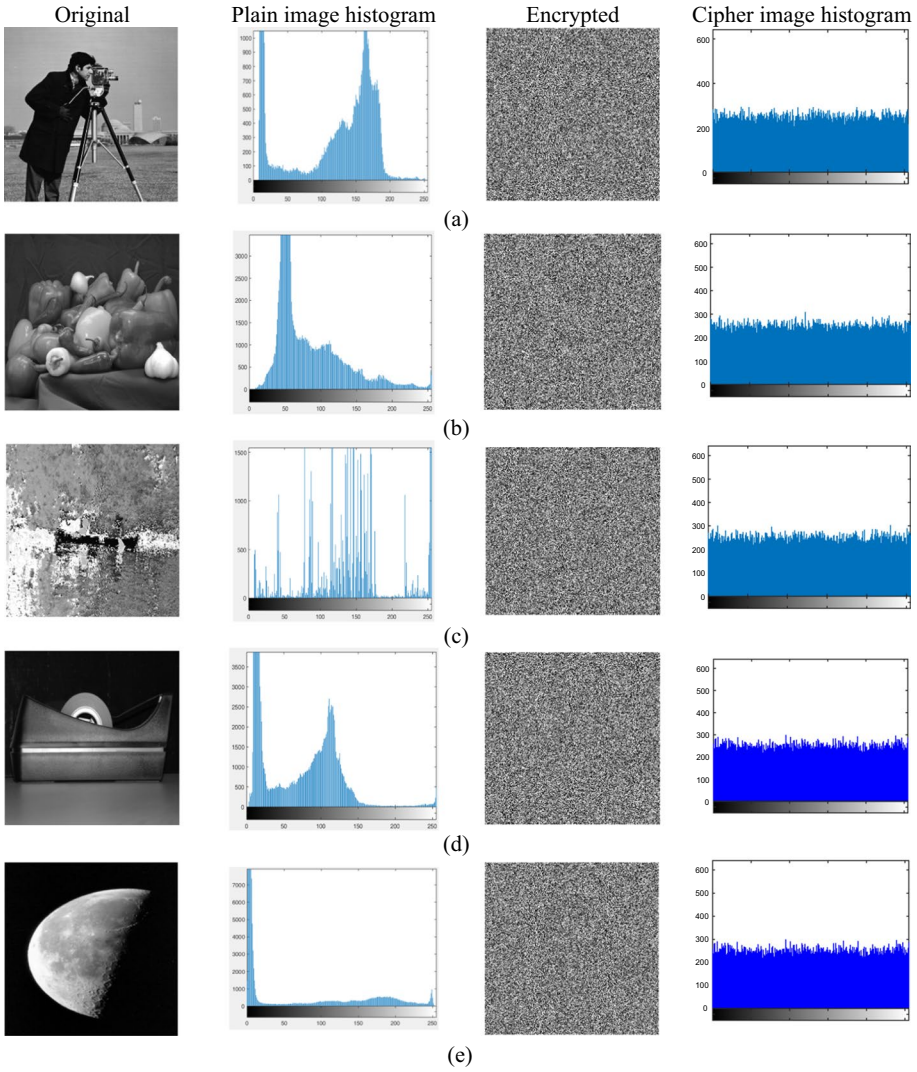


Fig. 5 Image histogram analysis for the test images: **a** Cameraman, **b** Peppers, **c** Canoe, **d** Tape, **e** Moon, **f** DICOM_1, **g** DICOM_2, **h** DICOM_3, **i** DICOM_4, **j** DICOM_5

4.1.3 Statistical Attack Analysis

A successful image encryption algorithm should generate the cipher image whose neighbouring pixels should not provide any statistical clue to break the algorithm. Thus, the developed algorithm should be tested for resistance against statistical attack. This attack analysis can be validated by analyzing the cipher image's histogram, entropy and correlation coefficient.

(a) Image histogram analysis

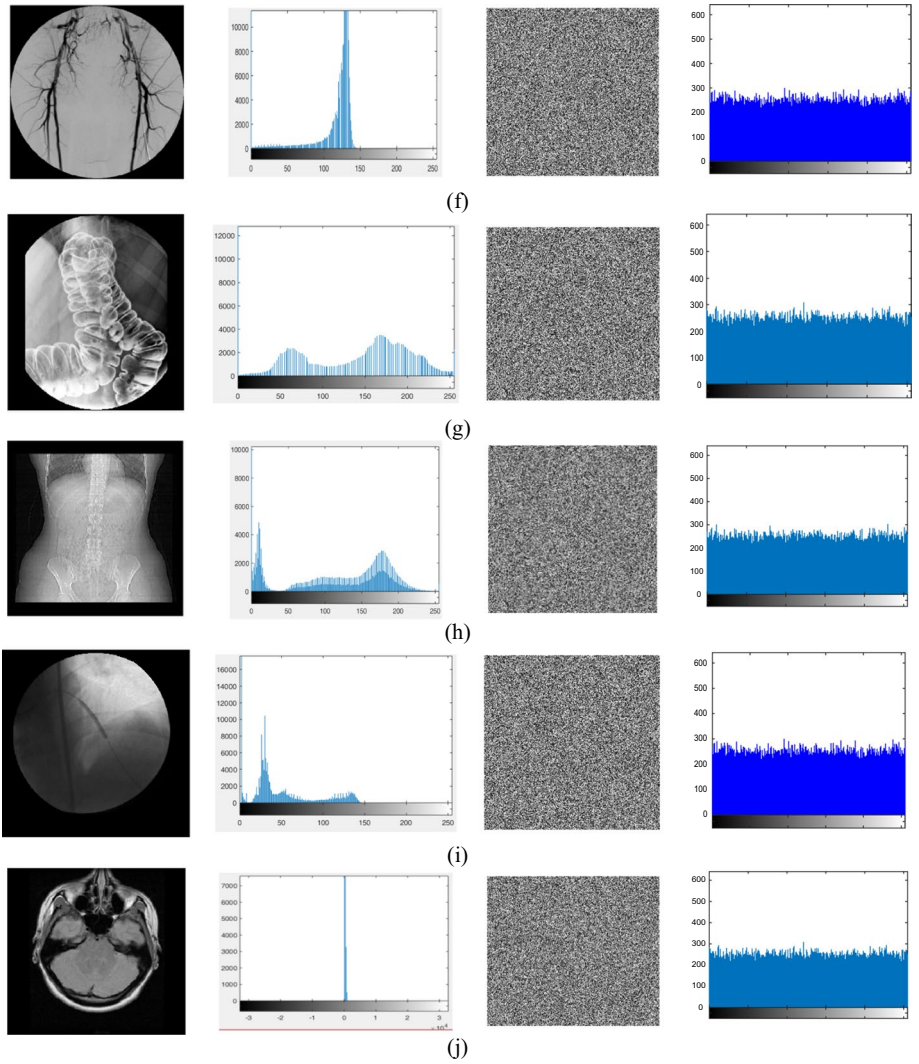


Fig. 5 (continued)

Image histogram is the graphical representation of pixels distribution in an image. It is defined as a plot between numbers of pixels occurring at each intensity level. For a random cipher image, the probability of occurrence of intensity levels should be uniform. Figure 5 shows the original and cipher image histograms for the test images. From Fig. 5, it can be seen that the histograms of the cipher image are flat, which means the pixels are uniformly

Table 5 Plain and cipher image entropy analysis

IMAGES	Plain image entropy	Cipher image entropy
Cameraman	7.0097	7.9974
Peppers	6.9837	7.9974
Canoe	7.4133	7.9972
Tape	6.7240	7.9964
Moon	5.4294	7.9950
DICOM_1	4.9310	7.9945
DICOM_2	5.5491	7.9974
DICOM_3	6.4002	7.9968
DICOM_4	4.7009	7.9937
DICOM_5	6.2387	7.9971

distributed at all intensities. Thus, this is evident that the statistical analysis is not possible for the proposed encryption algorithm.

(b) Entropy analysis

Shannon entropy is the standard metric to evaluate the randomness or uncertainty of the information. Information entropy can be mathematically represented as in Eq. (17),

$$\mathrm{H(m)} = -\sum_{(i=0)}^{L} \mathrm{P(mi)log_2P(mi)} \tag{17}$$

where m is the ith grey value for L level grey image, P(m) is the probability of m, so$\sum_{I=0}^{L}$ p(mi) = 1. The optimal information entropy value of an ideal cipher image with a range of [0, 255] is 8. If the entropy value is less than the optimal entropy value, then there exists some predictability to perform statistical attacks. Table 5 shows the information entropy analysis of the plain and cipher images. From Table 5, it is clear that the cipher image entropy values are closer to the optimal value. This proves that the cipher images developed from the proposed scheme are highly random to resist the statistical attacks.

(iii) Correlation analysis

This is one of the important analyses to study the statistical relationship between the neighbouring pixels of an image. The correlation coefficients of images are calculated using the following formula (18–20),

$$\mathrm{r_{xy}} = \frac{\mathrm{cov(x, y)}}{\sqrt{D(x)D(y)}} \tag{18}$$

$$\text{where, } \mathrm{D(x)} = \frac{1}{\mathrm{N}} \sum_{i=1}^{N} \left(x_i - E(x)\right)^2, \tag{19}$$

Table 6 Correlation analysis for the test images

Image		Horizontal	Vertical	Diagonal
Cameraman	Original	0.9311	0.9098	0.9098
	Encrypted	0.0347	0.0235	−0.0367
Peppers	Original	0.9849	0.9664	0.9764
	Encrypted	−0.0020	−0.0078	0.0382
Canoe	Original	0.7834	0.7431	0.7123
	Encrypted	0.0112	−0.0169	0.0071
Tape	Original	0.9924	0.9700	0.9715
	Encrypted	−0.0454	0.0217	0.0262
Moon	Original	0.9927	0.9852	0.9917
	Encrypted	0.0257	−0.0072	0.0148
DICOM_1	Original	0.9823	0.9883	0.9733
	Encrypted	0.0274	−0.0182	0.0266
DICOM_2	Original	0.9870	0.9931	0.9819
	Encrypted	0.0043	−0.0036	0.0013
DICOM_3	Original	0.9966	0.9950	0.9921
	Encrypted	0.0080	0.0107	−0.0005
DICOM_4	Original	0.9661	0.9756	0.9469
	Encrypted	0.0208	−0.0079	0.2180
DICOM_5	Original	0.9979	0.9957	0.9935
	Encrypted	−0.0214	0.0208	0.0057

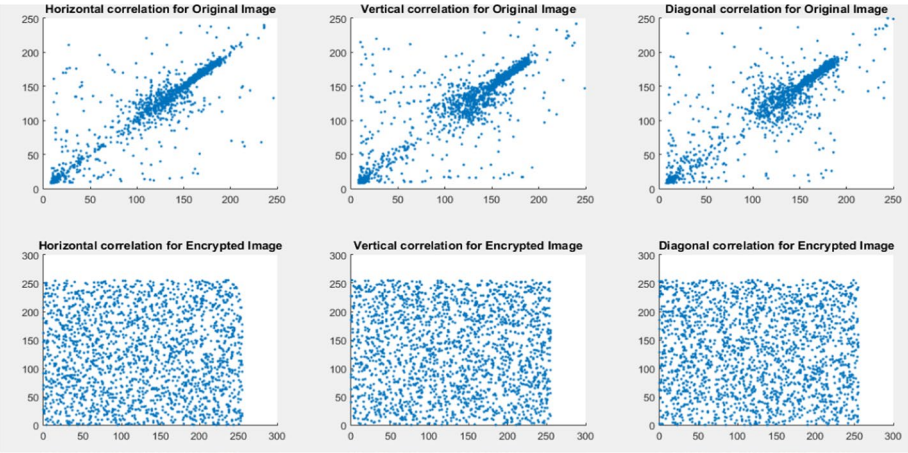


Fig. 6 Correlation Analysis in the horizontal, vertical and diagonal direction

Table 7 NPCR critical value analysis

Image	NPCR (%)	NPCR critical values		
		$N^{*}_{0.05} = 99.5693\%$	$N^{*}_{0.01} = 99.5527\%$	$N^{*}_{0.001} = 99.5341\%$
Cameraman	99.6192	Pass	Pass	Pass
Peppers	99.6406	Pass	Pass	Pass
Canoe	99.6253	Pass	Pass	Pass
Tape	99.6711	Pass	Pass	Pass
Moon	99.6986	Pass	Pass	Pass
DICOM_1	99.6299	Pass	Pass	Pass
DICOM_2	99.6604	Pass	Pass	Pass
DICOM_3	99.6330	Pass	Pass	Pass
DICOM_4	99.6513	Pass	Pass	Pass
DICOM_5	99.6131	Pass	Pass	Pass

Table 8 UACI critical value analysis

Image	UACI (%)	UACI Critical Values		
		$U^{*-}_{0.05} = 33.2824\%$ $U^{*+}_{0.05} = 33.6447\%$	$U^{*-}_{0.01} = 33.2255\%$ $U^{*+}_{0.01} = 33.7016\%$	$U^{*-}_{0.001} = 33.1594\%$ $U^{*+}_{0.001} = 33.7677\%$
Cameraman	33.2935	Pass	Pass	Pass
Peppers	33.5461	Pass	Pass	Pass
Canoe	33.3086	Pass	Pass	Pass
Tape	33.6181	Pass	Pass	Pass
Moon	33.5989	Pass	Pass	Pass
DICOM_1	33.3531	Pass	Pass	Pass
DICOM_2	33.5263	Pass	Pass	Pass
DICOM_3	33.4651	Pass	Pass	Pass
DICOM_4	33.2889	Pass	Pass	Pass
DICOM_5	33.4041	Pass	Pass	Pass

$$\mathrm{cov(x,y)} = \frac{1}{\mathrm{N}} \sum_{i=1}^{N} \big(x_i - E(x)\big) - \big((y_i - E(y)\big), \text{ and } \mathrm{E(x)} = \frac{1}{\mathrm{N}} \sum_{i=1}^{N} x_i \tag{20}$$

where x and y are the grey value of two adjacent pixels in the image, cov(x, y) is covariance, D(x) is variance and E(x) is mean. The statistical relationship between the adjacent pixels of an ideal cipher image should be completely broken. Thus, the correlation coefficient of an ideal cipher image should be nearer to zero to resist the statistical analysis. Correlation coefficients for the test images are computed in all the three directions, i.e., horizontal, vertical and diagonal, and the values are tabulated in Table 6. The results from Table 6 show that the correlation values of the cipher images in all the three directions are nearer to zero. Further, Fig. 6 plots the adjacent pixels in the intensity space [0, 255]. From Fig. 6, it is again proved that there is no correlation between the neighbouring pixels.

4.1.4 Differential Attack Analysis

This is another important analysis that the developed encryption algorithm should possess to defy the different attacks. Number of Pixel Change Rate (NPCR) and Unified Average Change in Intensity (UACI) are the two essential metrics in this analysis. To defend the differential attack, the value of NPCR and UACI should be in the critical range. The NPCR and UACI values of the test images and their critical value estimation are tabulated in Tables 7 and 8 respectively. From the table, it can be inferred that the NPCR and UACI values obtained from the proposed algorithm have successfully passed the critical values at the significance levels 1%, 5% and 10%.

4.2 Communication Channel Analysis

The original images are encrypted in the upper and physical layer of DCT precoded OFDM system with 16-QAM technique and transmitted over the Additive White Gaussian Noise (AWGN) channel. In this section, the performance of the proposed DCT precoded encryption and modulation scheme is evaluated with the AWGN channel. The quality of the deciphered and demodulated image at the received end is evaluated by Peak Signal to Noise Ratio (PSNR).PSNR can be computed using the mathematical expression as in Eq. (21),

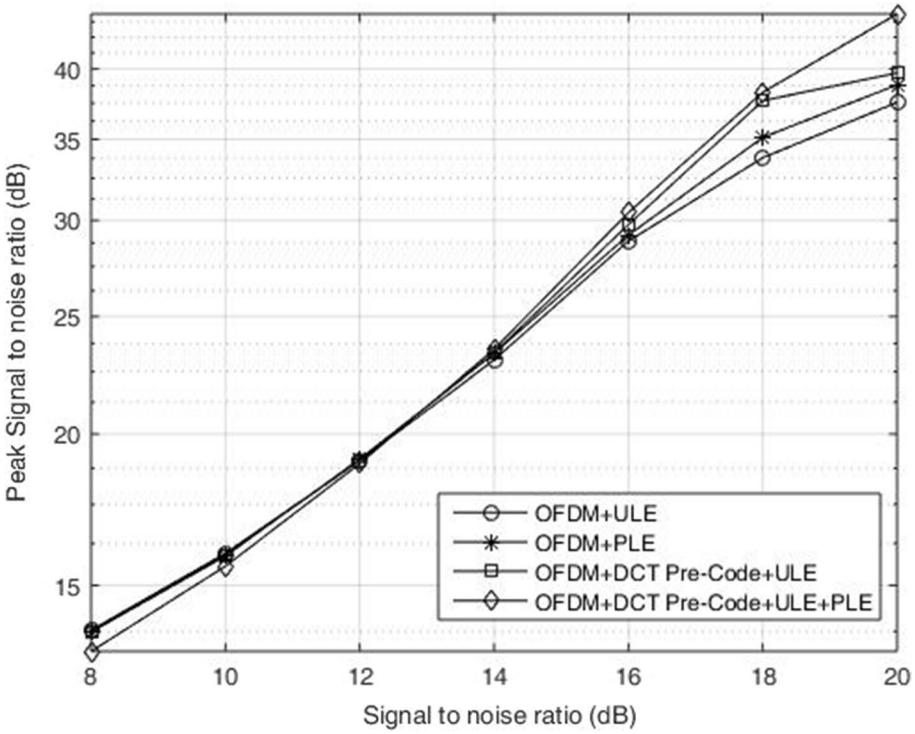


Fig. 7 PSNR Versus SNR plot of the deciphered image with AWGN channel

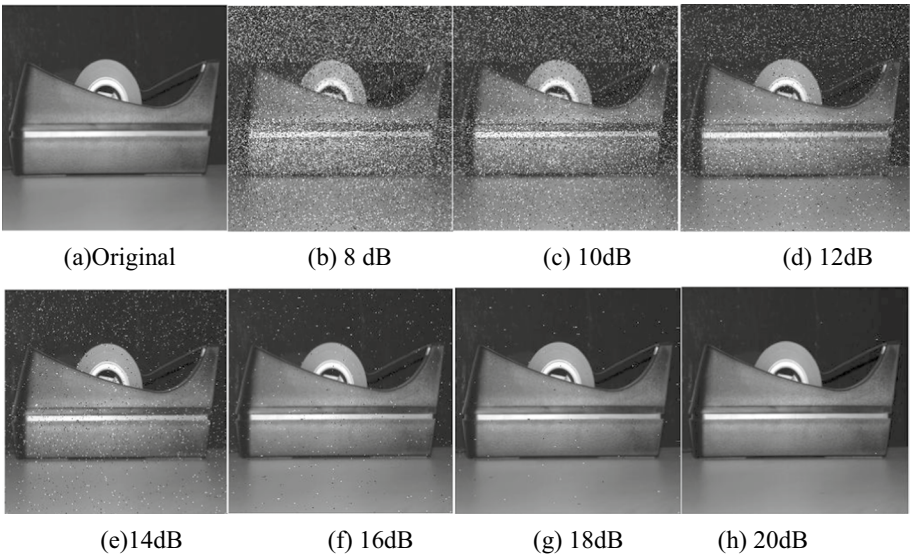


Fig. 8 Original and all decrypted images for 'Tape' image with varying SNR in AWGN channel

$$\text{PSNR} = 10log_{10}\frac{Imax^2}{MSE} \tag{21}$$

where Imax is the maximum intensity level of the image. For an 8-bit depth grey scale image, Imax = 2^8–1 = 255. Mean Square Error (MSE) is represented as in Eq. (22),

$$\text{MSE} = \frac{1}{M \times N}\sum_{i=1}^{M}\sum_{j=1}^{N}\left[P(i,j) - D(i,j)\right]^2 \tag{22}$$

where P(i, j) is the original plain image and D(i, j) is the deciphered image of size M × N. PSNR versus SNR plot is obtained for four different cases, (i) OFDM system only with ULE, (ii) OFDM system with PLE, (iii) OFDM system in the presence of DCT precoding and ULE, (iv) OFDM system with DCT precoding, ULE and PLE are shown in Fig. 7. From Fig. 7, it can be seen that the PSNR for case (iv) is better than all other cases. This is because the proposed chaotic interleaving technique as the physical layer encryption has reduced the burst errors happening with the AWGN channel. This scrambling, along with the DCT preceding, has improved the PSNR of the cipher images. Thus, the security of the QAM modulated DCT precoded OFDM system is enhanced when PLE, i.e., chaotic interleaving is added as a part of the system.

Further, the decrypted images at different SNRs for the test image are shown in Fig. 8. With SNR = 16 dB, the system under case (iv) has achieved PSNR > 30 dB. The result is greater than Wang et al.'s method, where PSNR ≈ 23 dB at SNR = 16 dB [17]. This proves that the system has achieved enhanced security and good visual quality of the decrypted image.

Table 9 PAPR analysis of the proposed system design with four different cases

Image	PAPR for case (i) (dB)	PAPR for case (ii) (dB)	PAPR for case (iii) (dB)	PAPR for case (iv) (dB)
Cameraman	13.7030	8.2518	11.4199	7.0821
Peppers	13.6628	8.2348	11.3967	7.7442
Canoe	13.6429	7.5251	10.4136	6.1588
Tape	13.1552	6.7593	9.3507	5.9493
Moon	12.7499	7.8591	10.8751	6.4092
DICOM_1	12.8426	7.6203	10.5459	7.0169
DICOM_2	13.7091	7.4451	10.3024	6.2241
DICOM_3	13.5366	7.7319	10.6991	6.9659
DICOM_4	12.8484	7.7386	10.7099	6.7528
DICOM_5	14.2118	7.5065	10.3872	6.0338

Case (i): ULE + PLE + Without Clipping

Case (ii): ULE + PLE + CR = 12

Case (iii): ULE + PLE + DCT Pre-code + Without clipping

Case (iv): ULE + PLE + DCT Pre-code + CR = 12

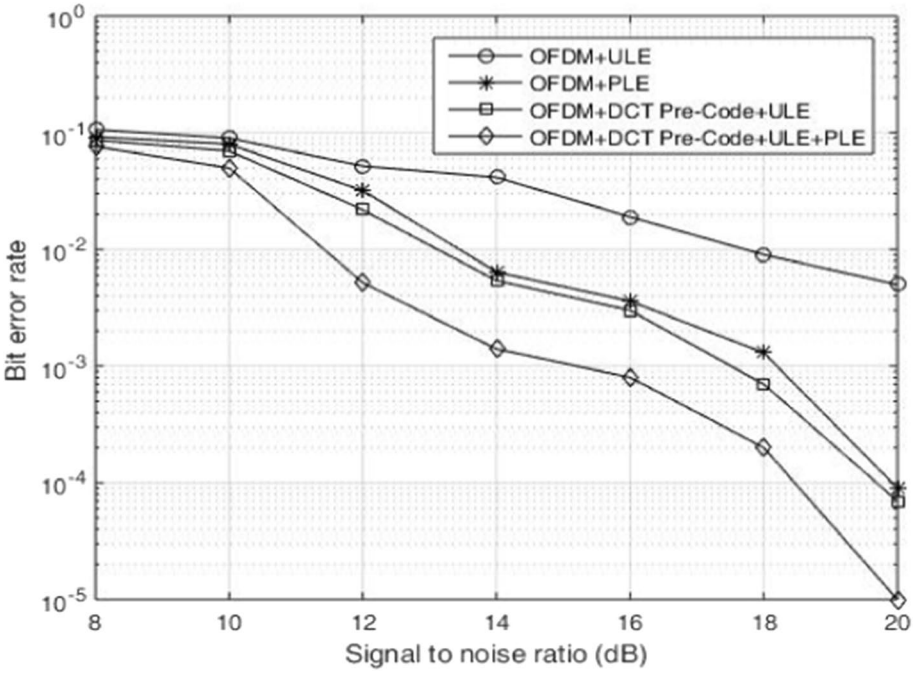


Fig. 9 BER Versus SNR plot

Table 10 Comparisons between the proposed method and the existing method

Ref.	Key space	PAPR	BER
Proposed work	10^{126}	6.63	10^{-5}
Wei et al. [28]	10^{72}	14.5	10^{-4}
Wu et al. [27]	10^{135}	7.9	10^{-4}
Wu et al. [29]	10^{91}	10.5	10^{-5}
Liu et al. [25]	10^{111}	-	10^{-4}

4.3 PAPR Analysis

This is one of the essential analyses used to evaluate the performance of OFDM system. Several experiments are carried out in the literature for PAPR reduction on image transmission in the OFDM system. This section evaluates the PAPR of the encrypted image in the proposed DCT precoded OFDM system with and without clipping. Let X_n be the discrete input signal of the OFDM system which is the bitstreams of the cipher image from upper layer encryption. The PAPR is the ratio of peak power and average power of the discrete OFDM signal. The analysis from [30] shows that the influence of clipping may improve the power efficiency, but on the other side, it may affect the Bit Error Rate (BER) performance of a system. From their experiments with the multipath fading channel, they have concluded that the Clipping Ratio (CR) higher than 6 dB may provide the trade-off between PAPR performance and BER performance. PAPR for different cases with and without clipping are tabulated in Table 9. From Table 9, it can be understood that the OFDM system with CR = 12, upper and physical layer encryption provides better PAPR reduction for the cipher images. The obtained results are better than [17] whose PAPR value is greater than 10 dB.

4.4 BER Analysis

BER performance of the developed secure DCT precoded OFDM system is evaluated in this section. BER is evaluated under different design cases with the same amplitude constraints. Same amplitude constraints mean the same clipping ratio. As discussed in the previous section, the CR is maintained to 12 for all the design cases. BER versus SNR plot is obtained for four different cases, and the plot is shown in Fig. 9. From Fig. 9, it can be seen that for the case (iv), i.e., OFDM with DCT pre-code and PLE has achieved very less BER for high SNR. At SNR = 16 dB, BER = 10^{-2} for OFDM with PLE and OFDM with DCT pre-code. If both PLE and DCT pre-code are combined in the system design, BER has improved to 10^{-3} at SNR = 16 dB. At SNR = 20 dB, BER is very much less 10^{-5} for case (iv), and this proves the efficiency of the proposed system.

Further, the performance of the proposed system is compared with the state-of-art methods available in the literature. Comparison is carried out in terms of key space, PAPR and BER and the results are tabulated in Table 10.

5 Conclusion

In this paper, a hybrid image encryption model based on combined chaotic system has been proposed. Bit level and symbol level encryption has carried out in the upper and physical layer of the DCT precoded OFDM system. Encryption quality of the proposed system has proved to be increased with the use of CCM and DNA coding. The system has achieved enhanced security with the key space 10^{126} and proved to be immune against statistical and differential attacks. With the use of DCT precoding and PLE, the PAPR of the system is reduced to 6 dB with CR = 12 and BER is very much reduced to 10^{-5}. Thus, the proposed system design can be adapted efficiently for real time secure wireless communication systems.

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Data availability All data generated or analysed during this study are included in this published article.

Declarations

Conflict of interest The authors do not have conflicts of interest associated with this publication.

Ethical approval This article does not contain any studies with human participants or animals performed by any authors.

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